Post-Training for Foundation Models

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Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - From specialists to generalists: birth of “foundation” models
  - Generalizing to any task: the art of prompting
  - Thinking slower: reasoning for better outcomes
  - Beyond text: prompting in other modalities
  - Prompting activity

- Assignments (Canvas):
  - Problem set 3 grades out (172 submissions: mean = 25.93 which is 103.7%, std dev: 1.87)
    - Email all regrade requests to our TA, Nick Cooper (e.g., a comment in Canvas is not sufficient)
    - Review session today at 2:30pm viz zoom (recording will be shared afterwards)
  - Lab assignment 3 due Friday
  - Final project starts after spring break

- Questions?

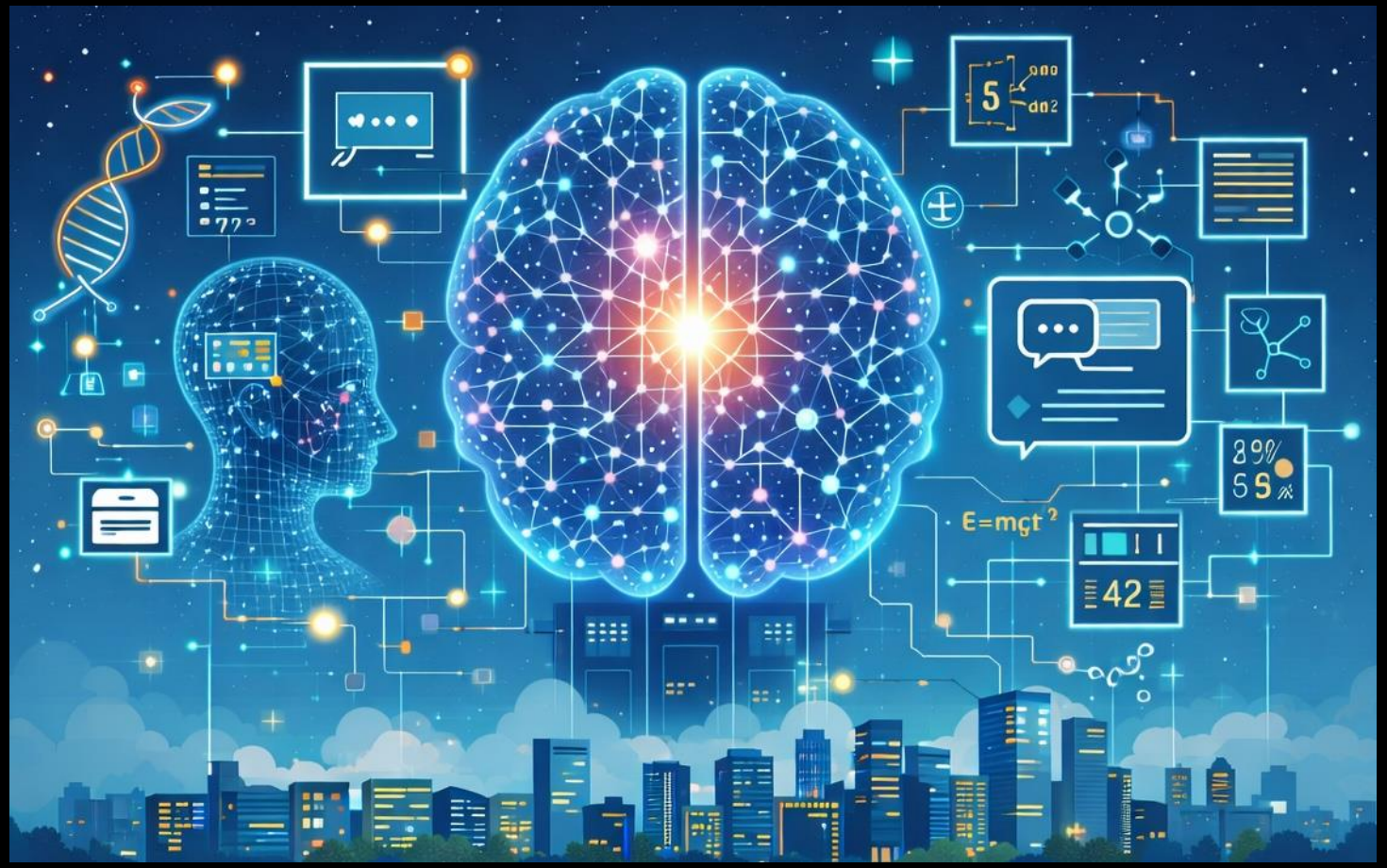
Outline

- The challenge: making foundation models useful
- Adapting models efficiently: parameter-efficient transfer learning
- Teaching models to follow human intent: instruction tuning
- Aligning models with human preferences: reinforcement learning from human feedback
- The shock: rapid societal adoption of foundation models

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Challenge: Harnessing Foundation Models' Intelligence for Real-World Impact



Massive Potential, Minimal Control: Why Raw Intelligence Needs Infrastructure (By Analogy)

New energy source that generates enormous amounts of electricity discovered

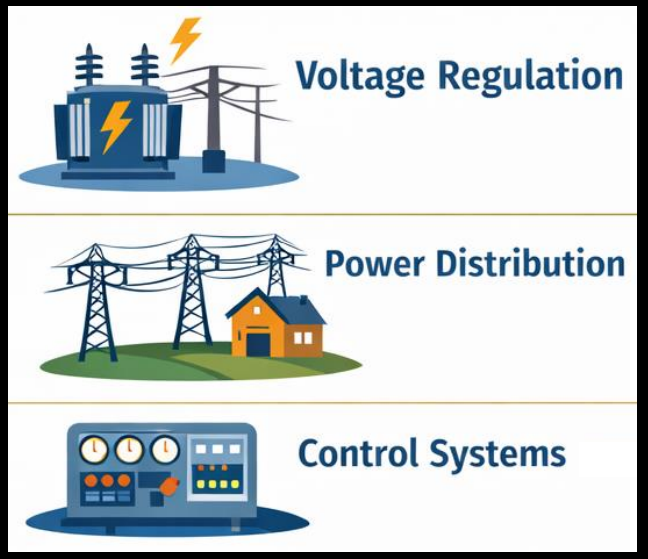


What's needed for this transition?

New power source benefits society



Infrastructure



Stabilizes irregular voltage flow

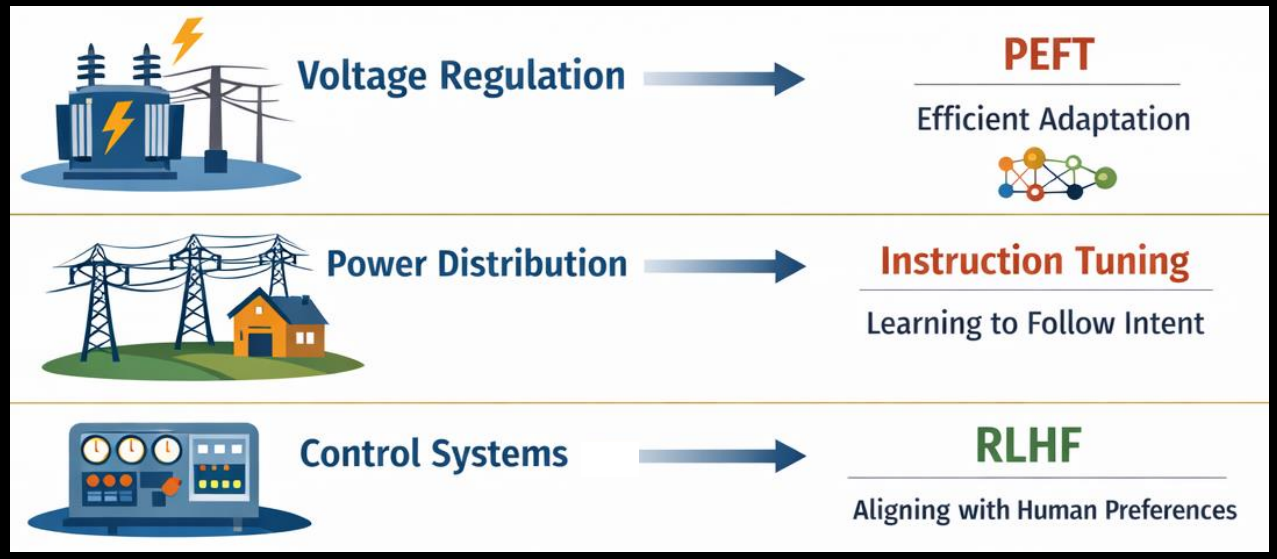
Delivers electricity from generator to users with transformers and power line

Manages electricity flow to ensure its safe, stable, and consistent

Massive Potential, Minimal Control: Why Raw Intelligence Needs Infrastructure (By Analogy)

Shaping how a model behaves with its acquired pretrained knowledge, typically over several stages rather than just with one fine-tuning step

Infrastructure



Outline

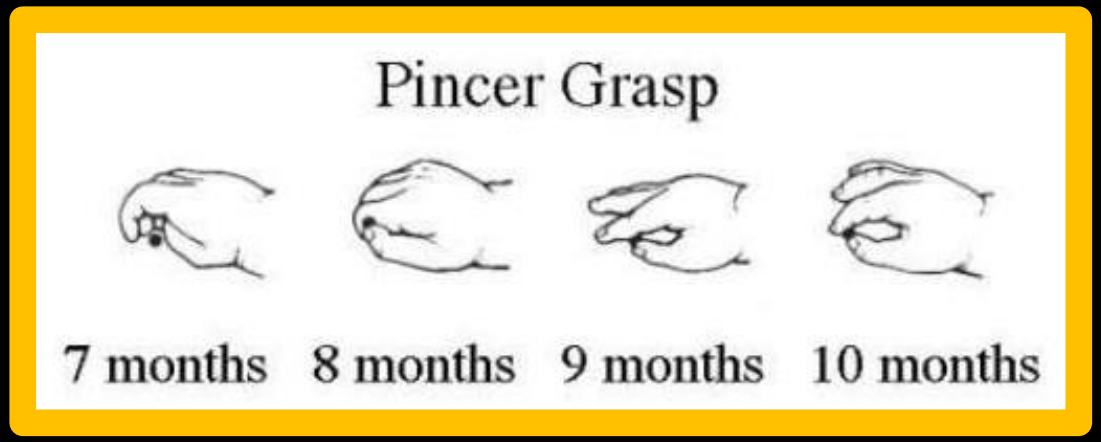
- The challenge: making foundation models useful
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Idea: Post-Training As Infrastructure



Tweak the output without modifying the model itself!

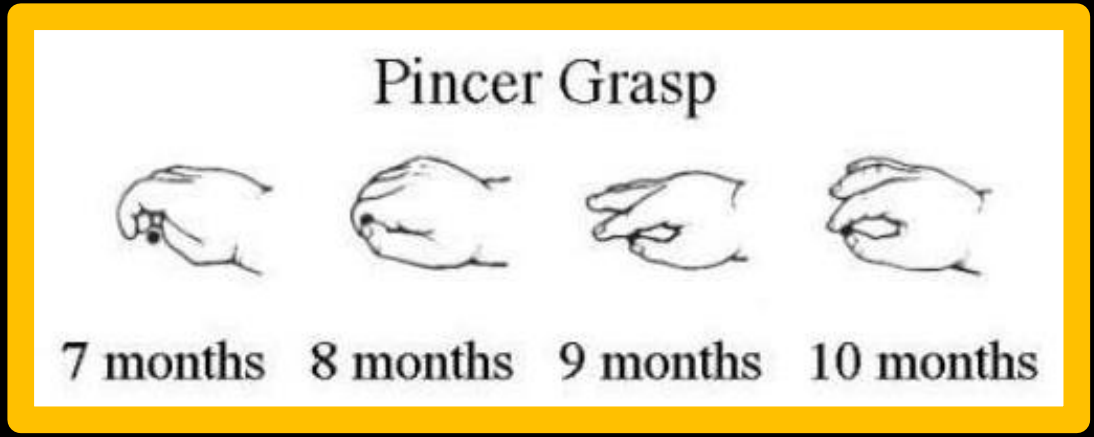
Recall: Models Learn Representations that Can be Reused for Many Tasks



https://www.facebook.com/1rvanisrisaravanan/posts/pincer-graspthe-pincer-grasp-is-the-coordination-of-the-index-finger-and-thumb-t/2833183393596559/



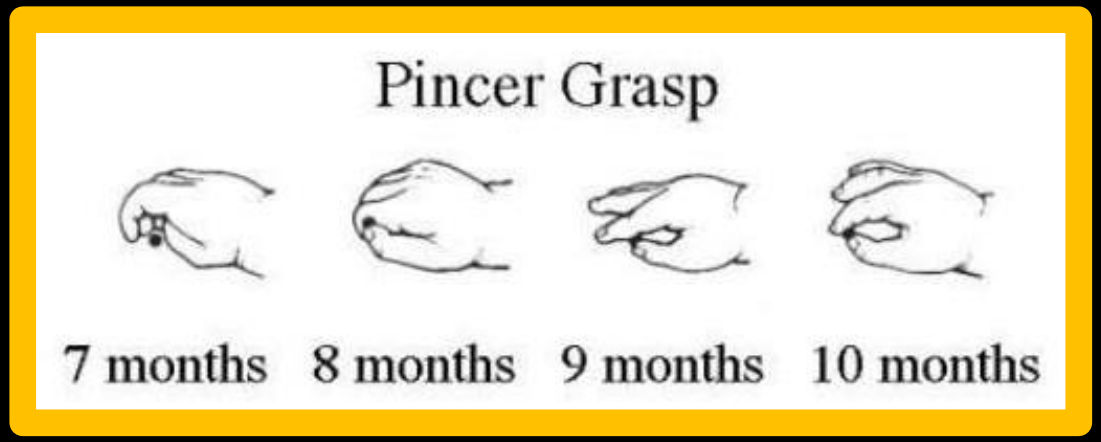
Why Not Just Fine-Tune the Foundation Model?



https://www.facebook.com/1rvanisrisaravanan/posts/pincer-graspthe-pincer-grasp-is-the-coordination-of-the-index-finger-and-thumb-t/2833183393596559/

- Catastrophic forgetting: knowledge lost when learning new tasks (noted by Michael McCloskey and Neal Cohen in 1989)
- Costly HW requirements: for 50B parameter model (32 bit precision) trained with Adam, need >1 TB of VRAM (GPU's memory)

Idea: Add Small Adaptor Modules that Help Models Perform Better for Downstream Tasks



https://www.facebook.com/1rvanisrisaravanan/posts/pincer-graspthe-pincer-grasp-is-the-coordination-of-the-index-finger-and-thumb-t/2833183393596559/







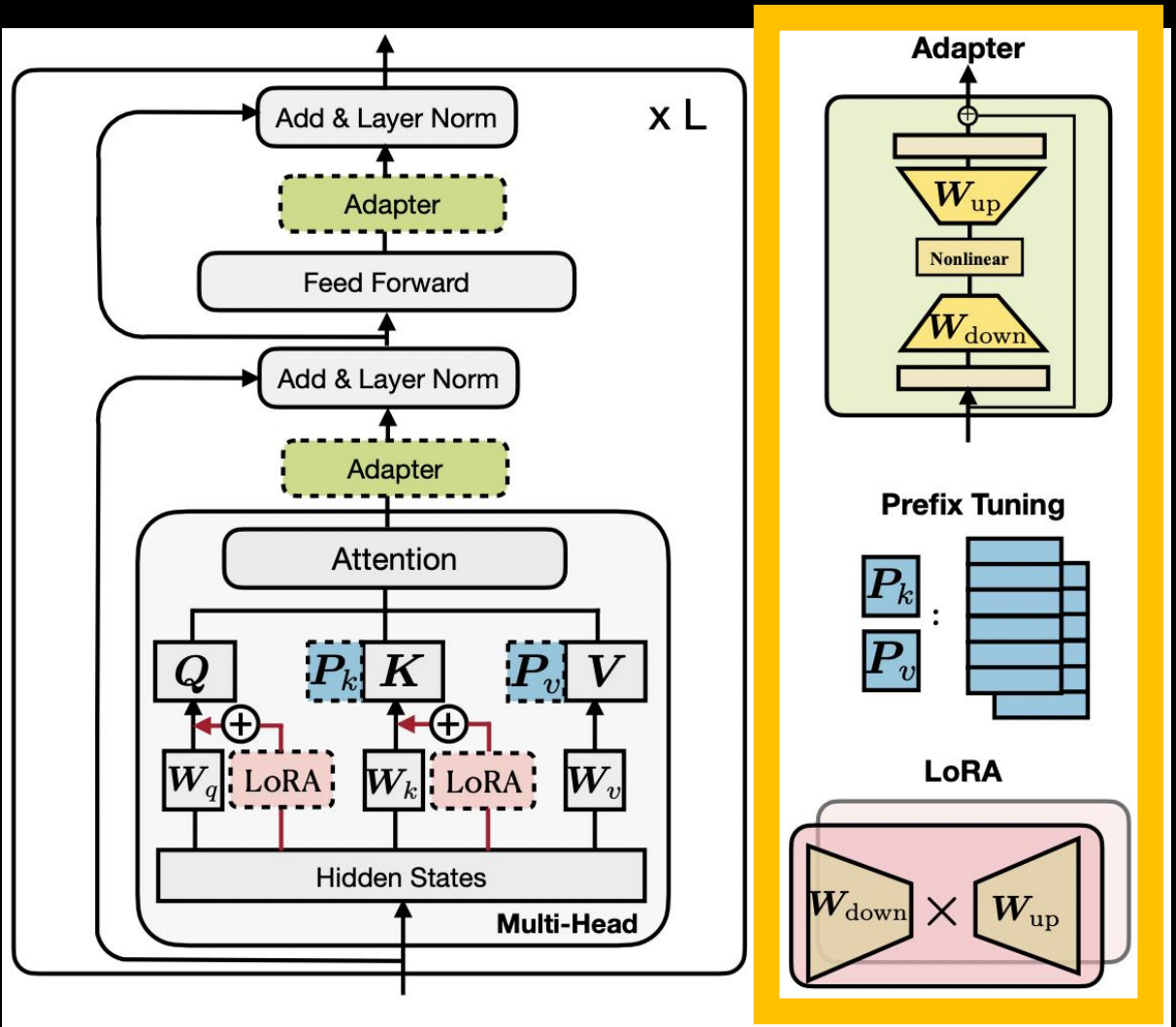


Figure 1: Illustration of the transformer architecture and several state-of-the-art parameter-efficient tuning methods. We use blocks with dashed borderlines to represent the added modules by those methods.

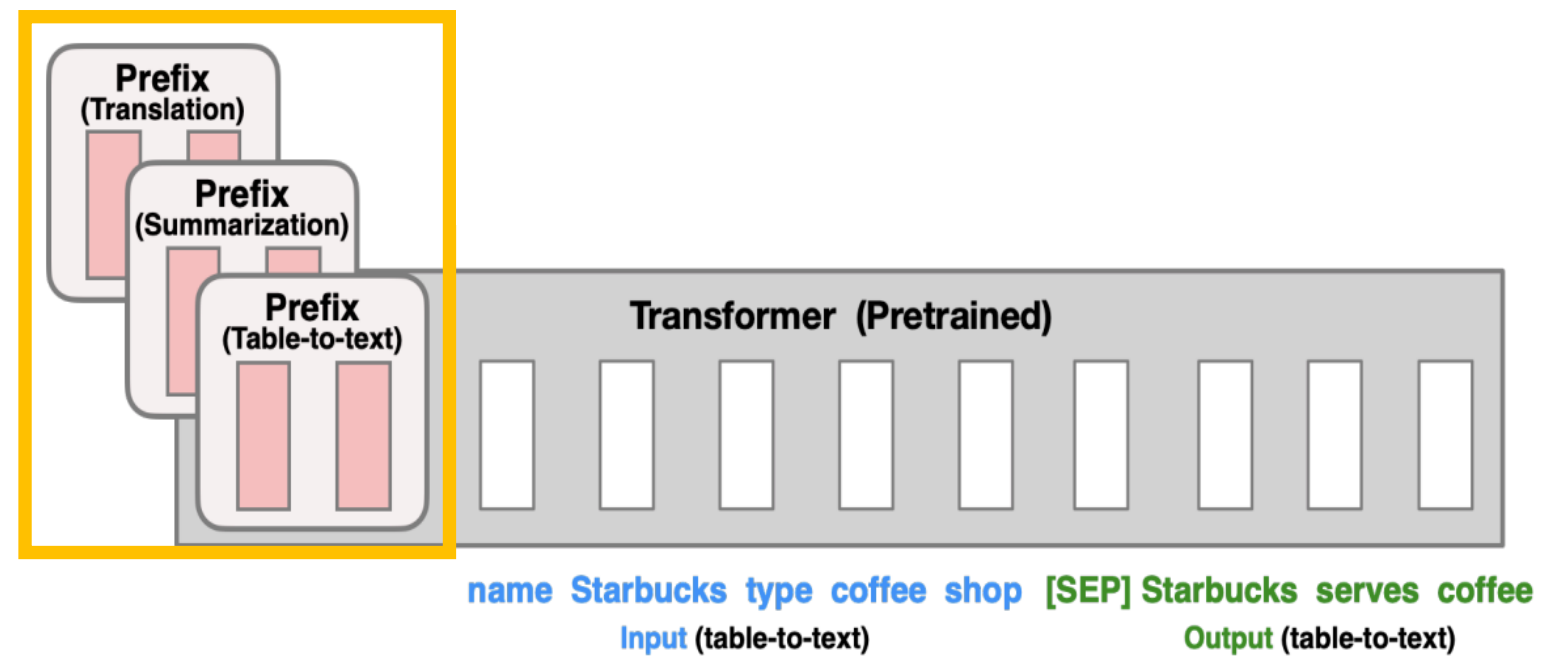
Approach:

Freeze Model and Train Only Added Parameters

(e.g., Three Key Types of Methods for Adding Parameters to Transformers)

He et al. Towards a Unified View of Parameter-Efficient Transfer Learning. ICLR 2022.

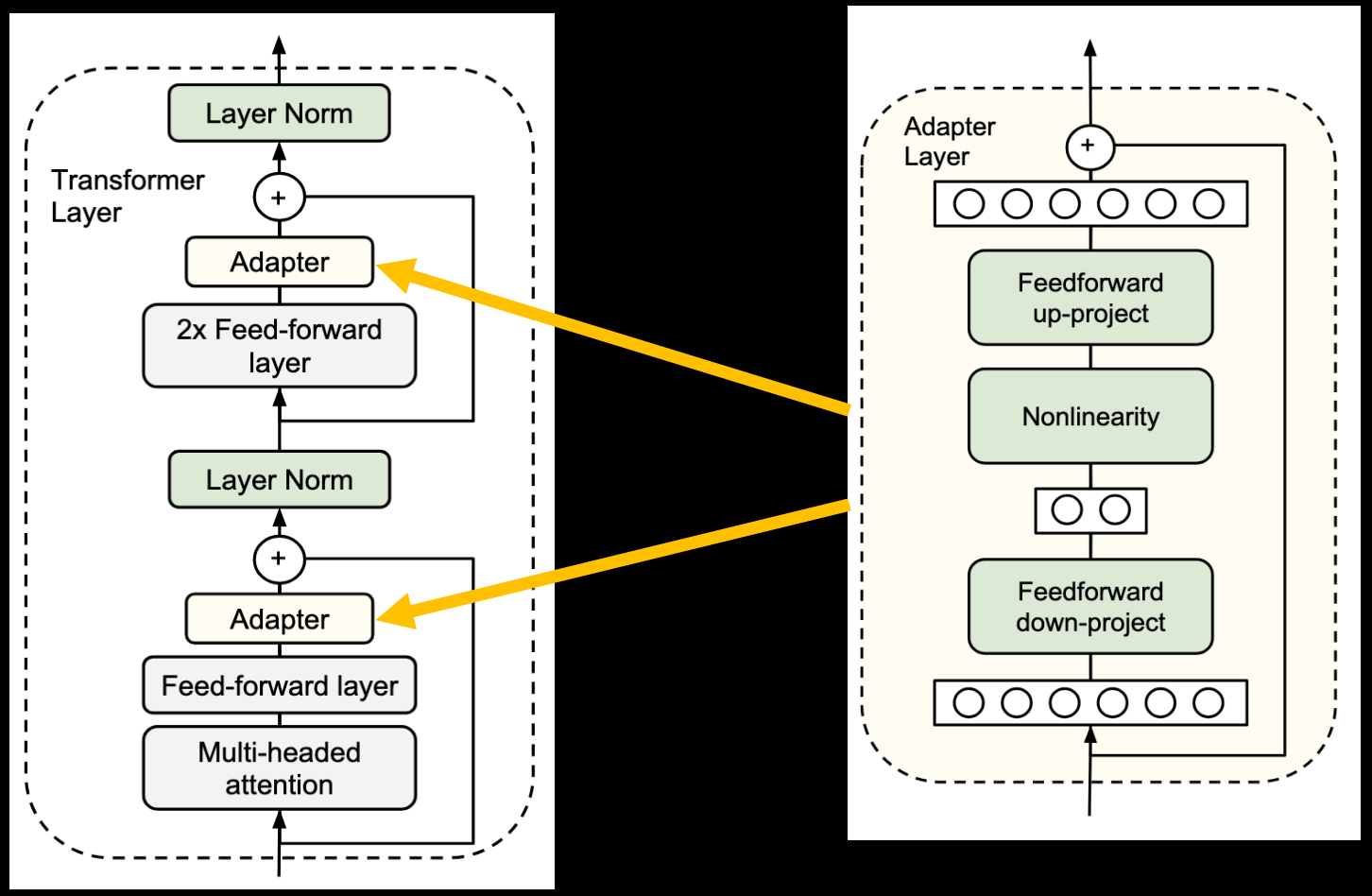
Prefix Tuning: Recall This Entails Training Extra Parameters as Continuous Prompts



Li and Liang. ACL 2021

How can the original model be recovered?

Adapter: Adds to Transformer Blocks Trainable Parameters (e.g., 3-5% More)

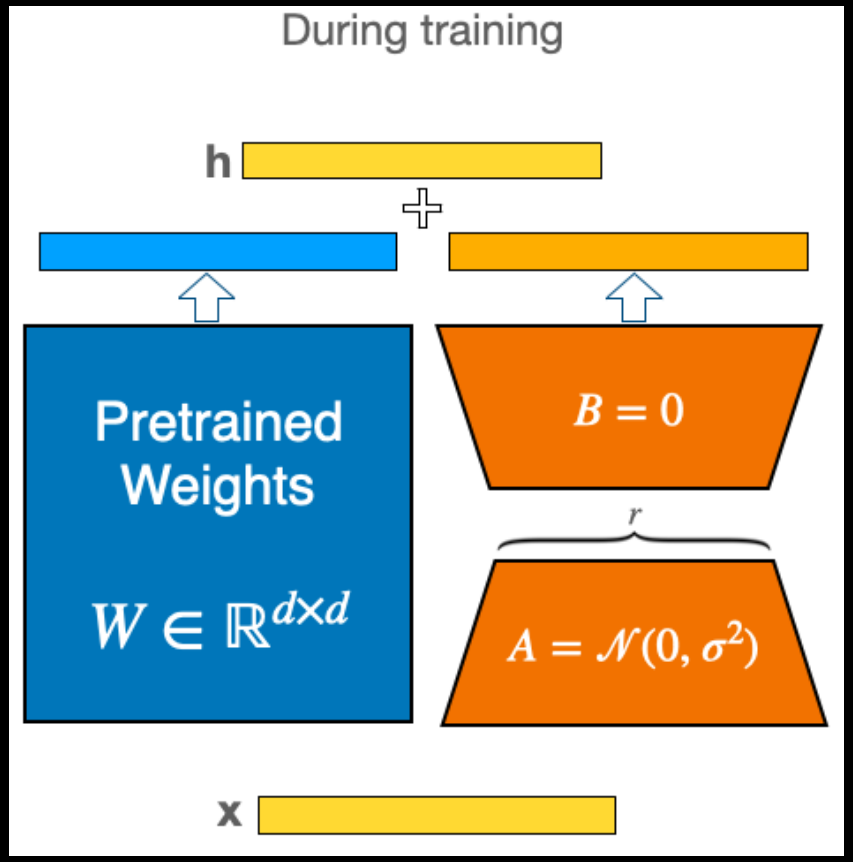


How can the original model be recovered?

[Houlsby, MLR19]

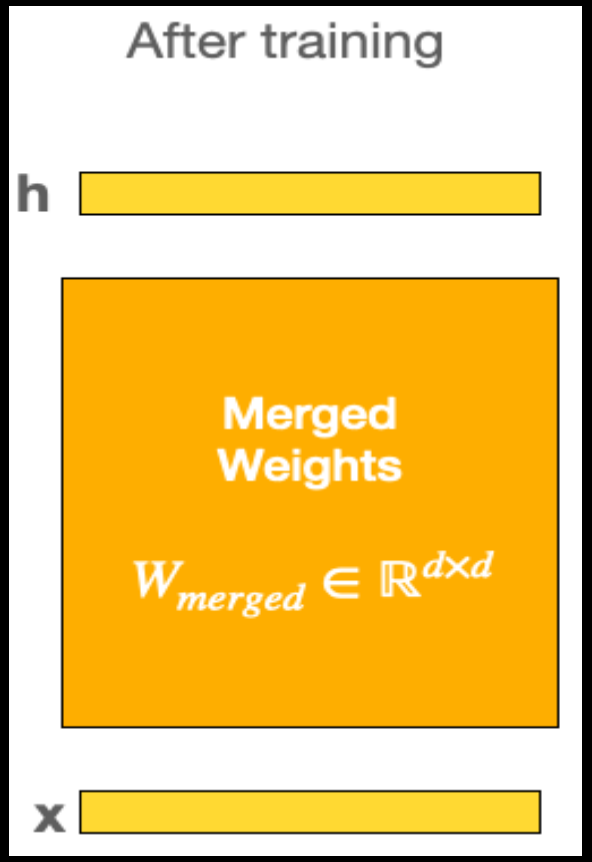
LoRA

How can the original model be recovered?



$$h = Wx + BAx$$

$$h = \underbrace{(W + BA)}_{W_{merged}}x$$



https://huggingface.co/docs/peft/main/en/conceptual_guides/lora

Training: trains low-rank matrices added to certain layers (typically attention), aka Low Rank Adaption

Inference: unlike prior two methods, no extra parameters and so compute/storage are needed because adapter weights are merged with original weights

[Hu, ICLR22]

PEFT Methods Achieve Competitive Performance with Full Fine-Tuning While Training 15% or Less Parameters

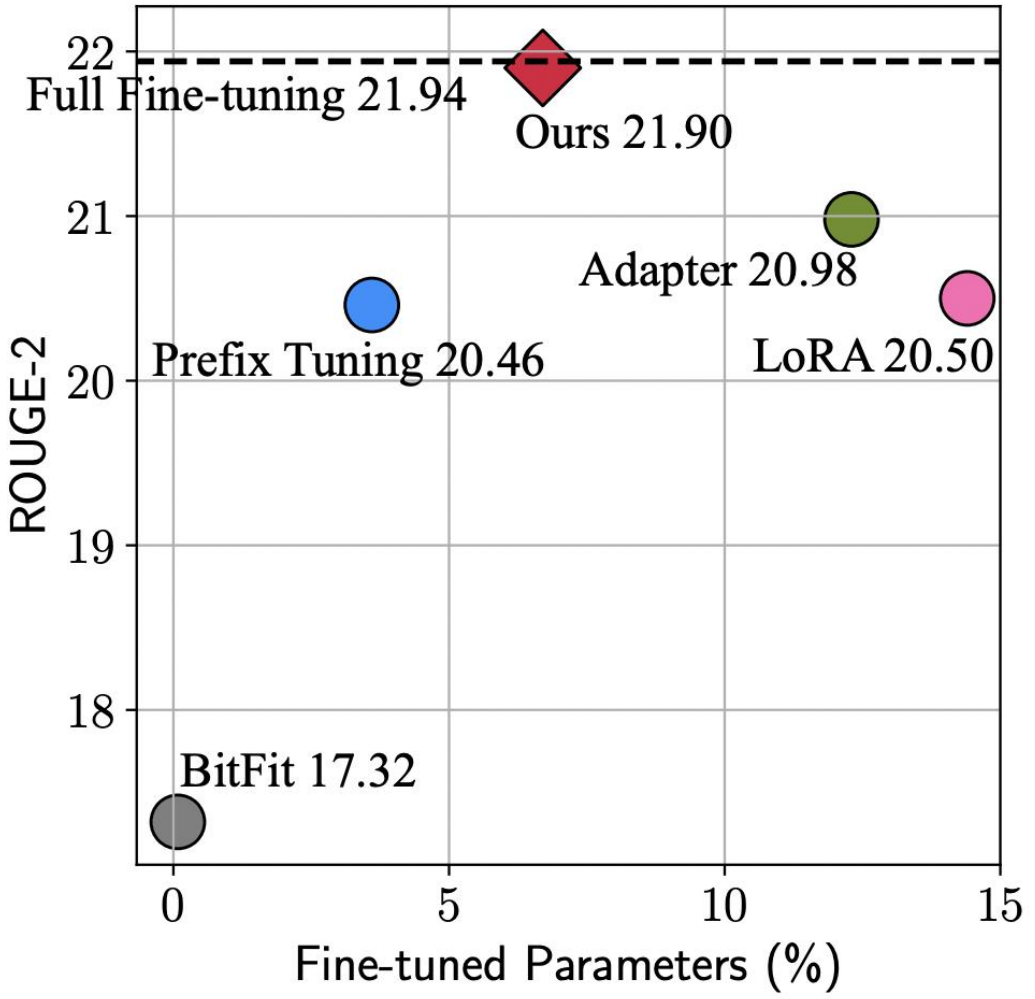


Figure 2: Performance of different methods on the XSum (Narayan et al., 2018) summarization task. The number of fine-tuned parameters is relative to the tuned parameters in full fine-tuning.

He et al. Towards a Unified View of Parameter-Efficient Transfer Learning. ICLR 2022.

PEFT Options — When to Prefer Each (Pre-Fix Fine Tuning, Adapter, LoRA)?

- Goal: inject task-specific context such as what style or personality to adopt when responding to a request.
  - Prefix tuning

- Goal: switch between related tasks easily such as text classification, summarization, and responding across many languages
  - Adapter

- Goal: perform better for a new target task
  - LoRA

- Note: LoRA (or a quantized variant QLoRA) is most common, and is regularly combined with other methods

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Idea: Post-Training As Infrastructure

Tweak the output to align with users' intentions!



(Coined the phrase “instruction tuning”)

Published as a conference paper at ICLR 2022

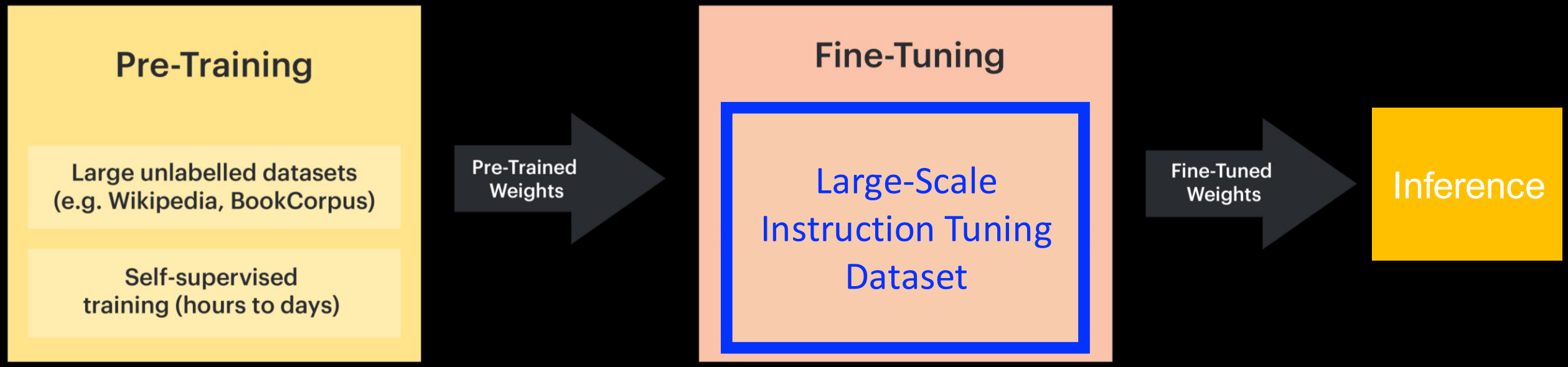
FINETUNED LANGUAGE MODELS ARE ZERO-SHOT LEARNERS

Jason Wei*, Maarten Bosma*, Vincent Y. Zhao*, Kelvin Guu*, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le

Google Research

“We show that instruction tuning—finetuning language models on a collection of datasets described via instructions—substantially improves zero-shot performance on unseen tasks.”

Approach



Default objective learned, such as predict next token

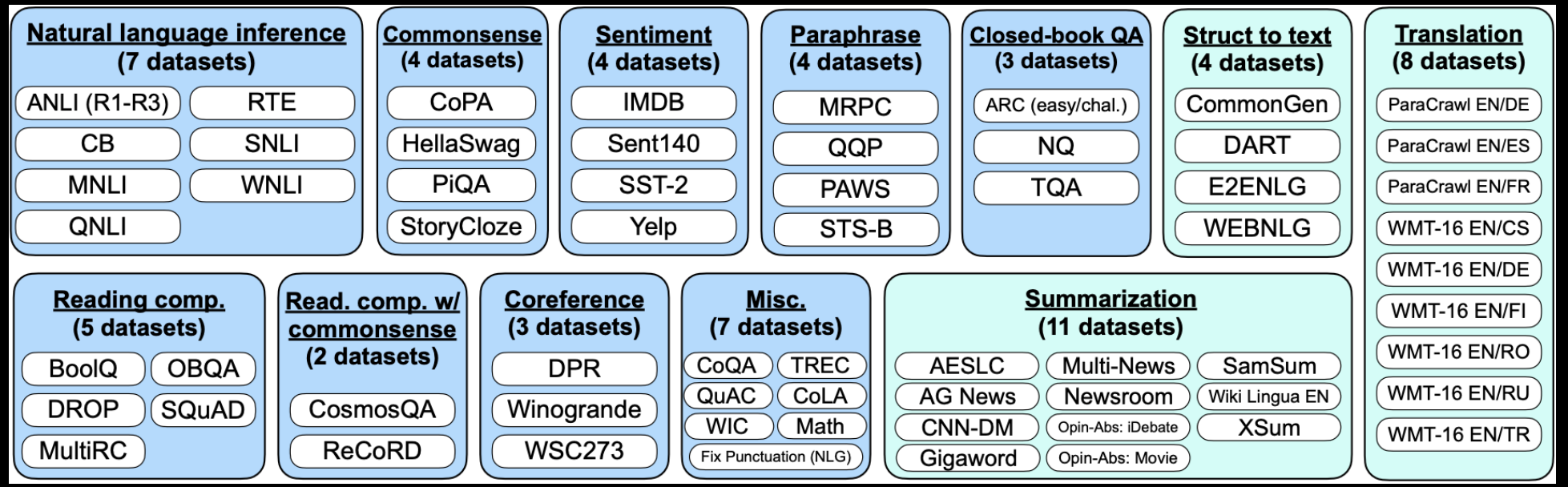
VS

Goal: respond well to users' prompts/instructions

https://docs.graphcore.ai/projects/bert-training/en/latest/bert.html

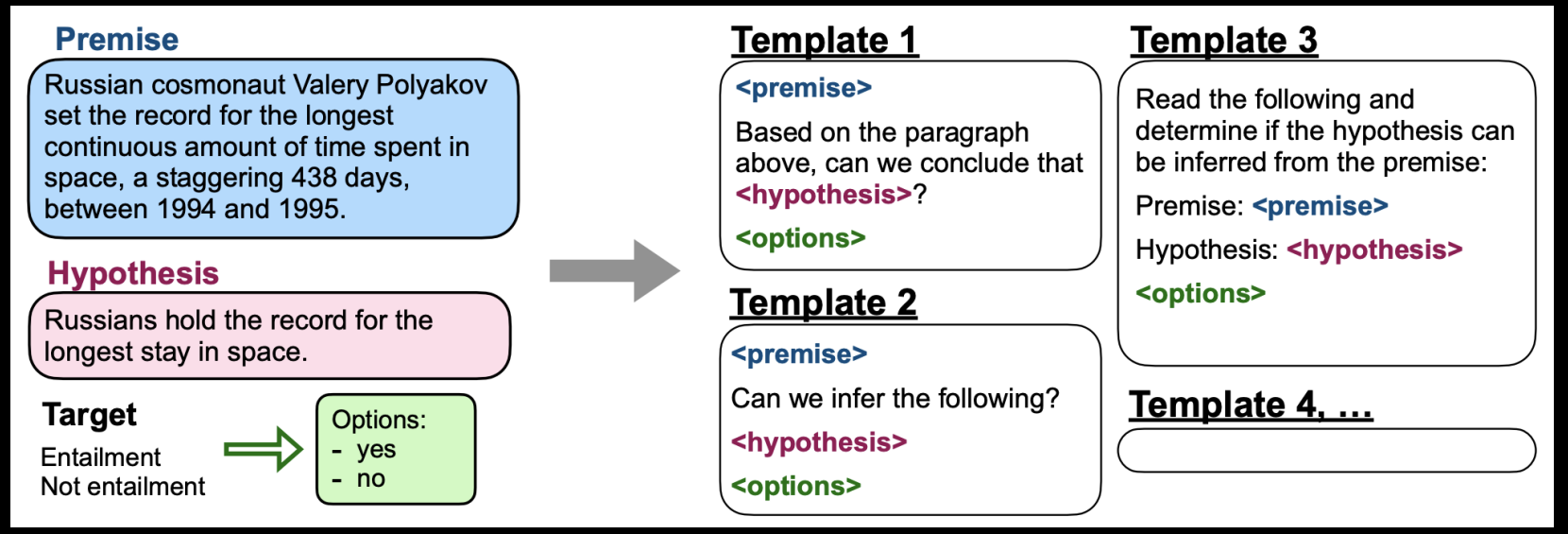
Approach: Creating Instruction Tuning Dataset

Adapts 62 text datasets to instructional formats:



10 hand-crafted templates per dataset of instructional formats anticipated for the target task at inference time:

At most 30k training examples generated per dataset to account for discrepancies in dataset sizes



[Wei et al., ICLR22]

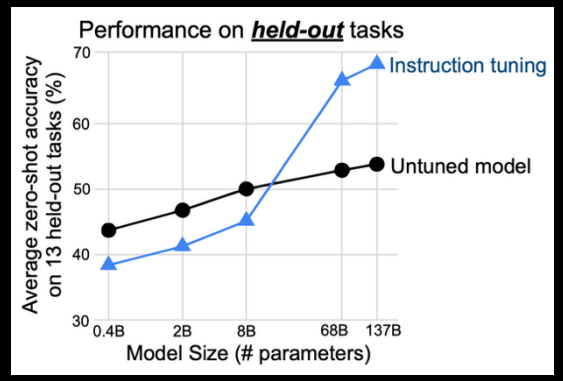
Examples: Creating Instructions

- Which represents generic SFT vs instruction tuning
  1. Input: Translate to French: "Good morning"
     Output: Bonjour
  2. Input: Write a professional email declining a meeting.
     Response: ...

- Which instruction do you think is better for training and why?
  - Explain blockchain
  - Explain blockchain in one sentence
  - Explain blockchain to a skeptical economist

Experiments

- Pretrained model: 137B parameter decoder-only transformer language model, trained on web documents, dialog data, and Wikipedia
- Fine-tuning: optimized with Adafactor (per-parameter method) for 30k gradient steps; took ~60 hours on a TPUv3
- Evaluation split: tested on held-out dataset clusters (zero-shot)
- Results:
  - Better zero-shot performance than 175B-parameter GPT-3 for 20 of 25 tested datasets
  - Better performance than GPT-3 used in few-shot settings for numerous datasets
  - Better performance when using continuous prompts
  - Performance gains are greater for larger models
  - Why might performance be worse for smaller models?



[Wei et al., ICLR22]

Instruction Tuning Also Benefits Multimodal Reasoning; e.g.,

Given: caption and bounding box object detections as descriptors of an image

Generate instruction tuning data: three types generated by GPT-4

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

[Liu et al., NeurIPS23]

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Idea: Post-Training As Infrastructure

Ensure generated outputs **align with human preferences**
(e.g., be truthful and do no harm)



(NeurIPS 2017)

Deep Reinforcement Learning from Human Preferences

Paul F Christiano
OpenAI
paul@openai.com

Jan Leike
DeepMind
leike@google.com

Tom B Brown
Google Brain*
tombbrown@google.com

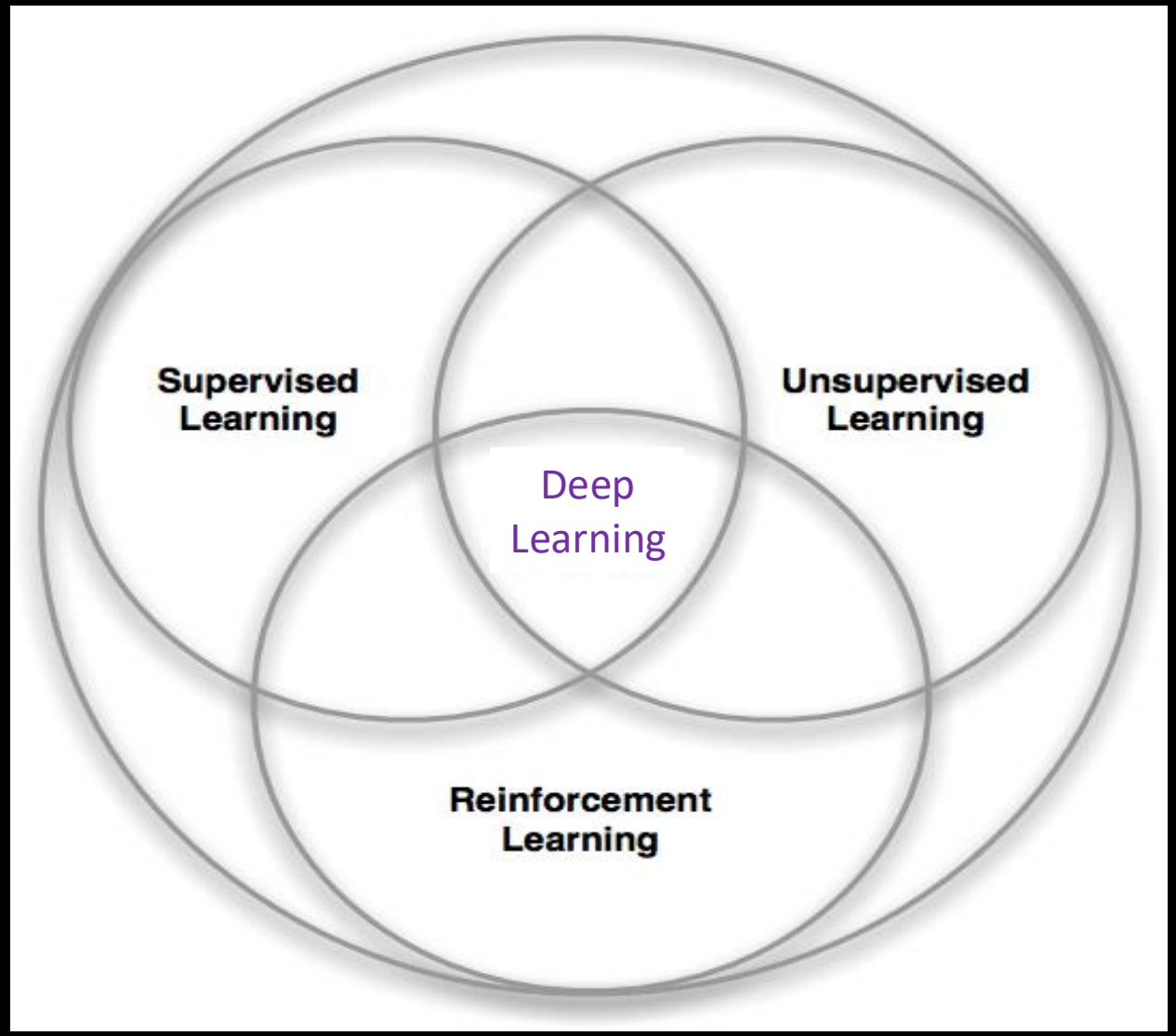
Miljan Martic
DeepMind
miljanm@google.com

Shane Legg
DeepMind
legg@google.com

Dario Amodei
OpenAI
damodei@openai.com

"Recent success in scaling reinforcement learning (RL) to large problems has been driven in domains that have a well-specified reward function… Unfortunately, many tasks involve goals that are complex, poorly-defined, or hard to specify… An alternative approach is to allow a human to provide feedback on our system's current behavior and to use this feedback to define the task… but using human feedback directly as a reward function is prohibitively expensive for RL systems that require hundreds or thousands of hours of experience… We overcome this difficulty by asking humans to compare possible trajectories of the agent, using that data to learn a reward function, and optimizing the learned reward function with RL…"

Background: Reinforcement Learning



Unsupervised learning involves least amount of human effort followed by, **supervised learning** the most with labels, and **reinforcement learning** strikes a middle ground with rewards

http://www0.cs.ucl.ac.uk/staff/d.silver/web/Teaching_files/intro_RL.pdf

Motivation: Less Human Supervision, Including for When Explicit Instruction is Impractical

- e.g.,



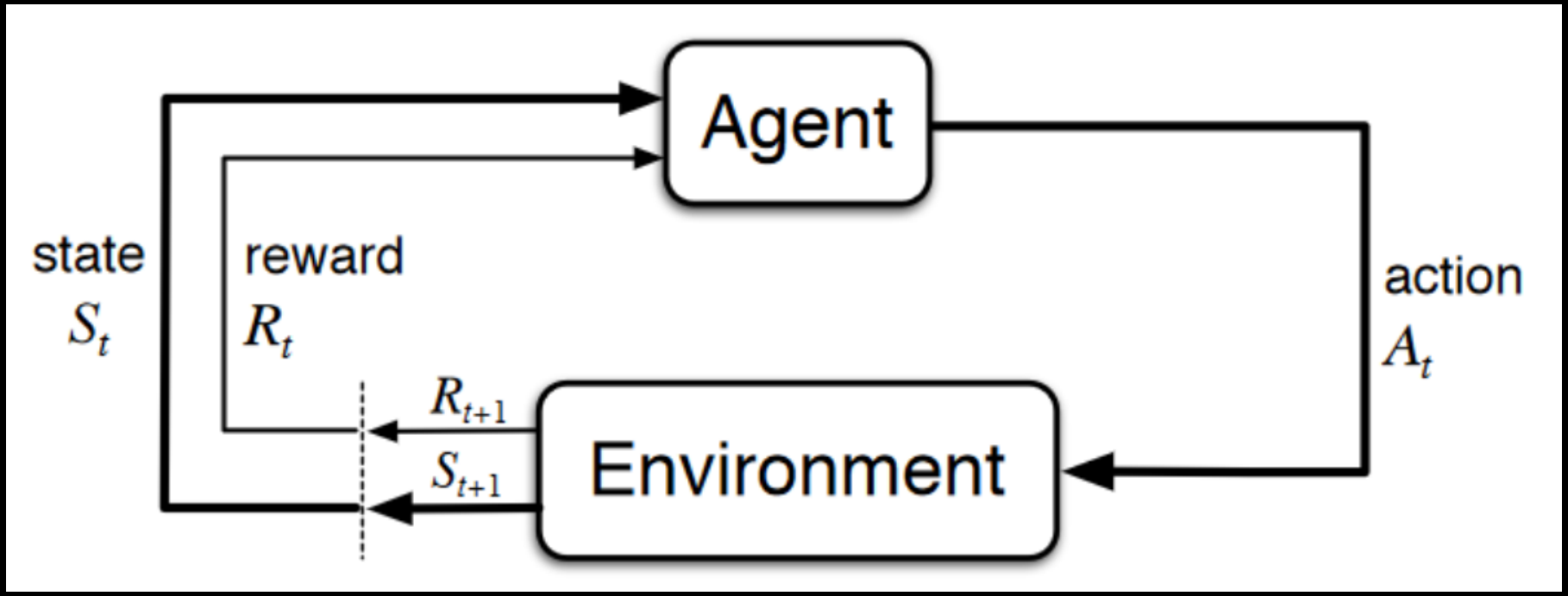
VS



- Let's try it! Please find what I want without me providing detailed instructions
  - Need: 1 volunteer
  - Your goal: Find the location in the room I am thinking of!
  - Rewards: I will indicate hot (near the target), cold (further from the target), and found after each ~5 seconds

Approach

Agent takes actions in an environment to maximize the total reward



https://towardsdatascience.com/applications-of-reinforcement-learning-in-real-world-1a94955bcd12

Approach by Example



https://www.youtube.com/watch?v=W_gxLKSsSIE&list=PL5nBAYUyJTrM48dViibyi68urttMlUv7e

Theoretical Foundation of RL: Markov Decision Processes (MDP)

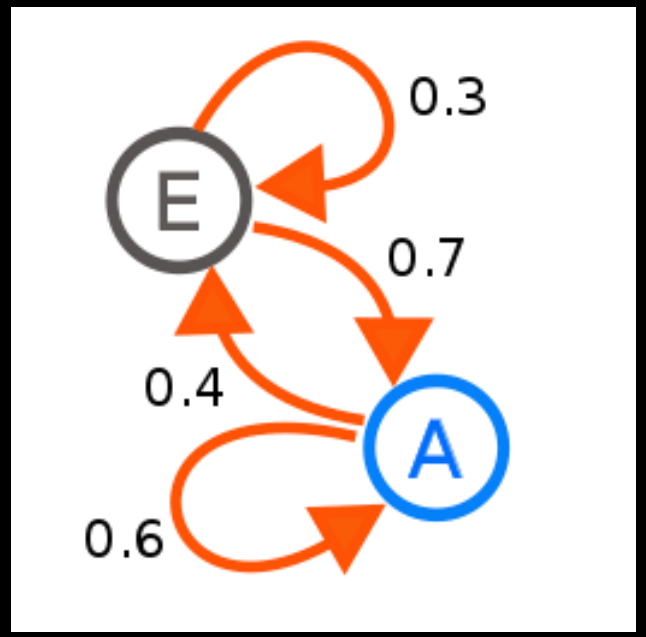
MDP consists of:

Markov process + Markov reward process + Actions

Markov Process (aka – Markov Chain)

- A set of states with transition probabilities defining system dynamics

e.g.,



  - How many states are in the above example?
- Markov property: only current state dictates future system dynamics

https://en.wikipedia.org/wiki/Markov_chain

Theoretical Foundation of RL: Markov Decision Processes (MDP)

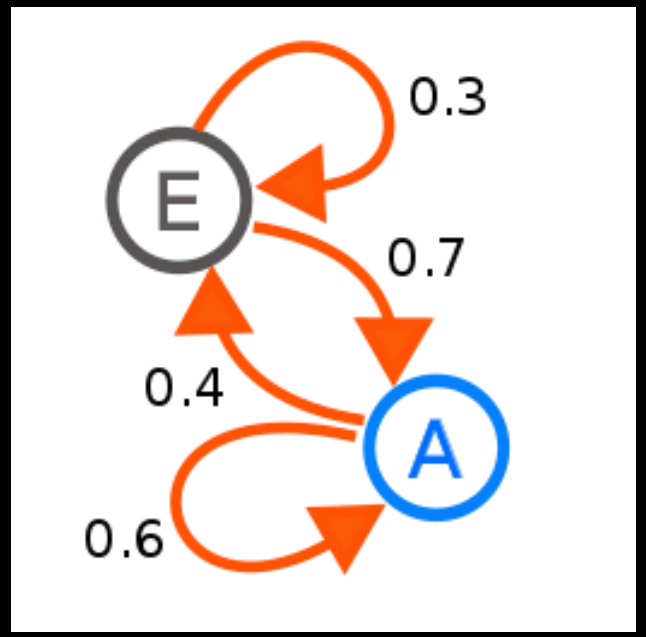
MDP consists of:

Markov process + Markov reward process + Actions

Markov Reward Process

- Additional variable accumulated over time to reflect the reward

e.g.,



- A discount factor between 0 and 1, gamma, indicates how far in the future rewards are considered to estimate a state's expected reward

https://en.wikipedia.org/wiki/Markov_chain

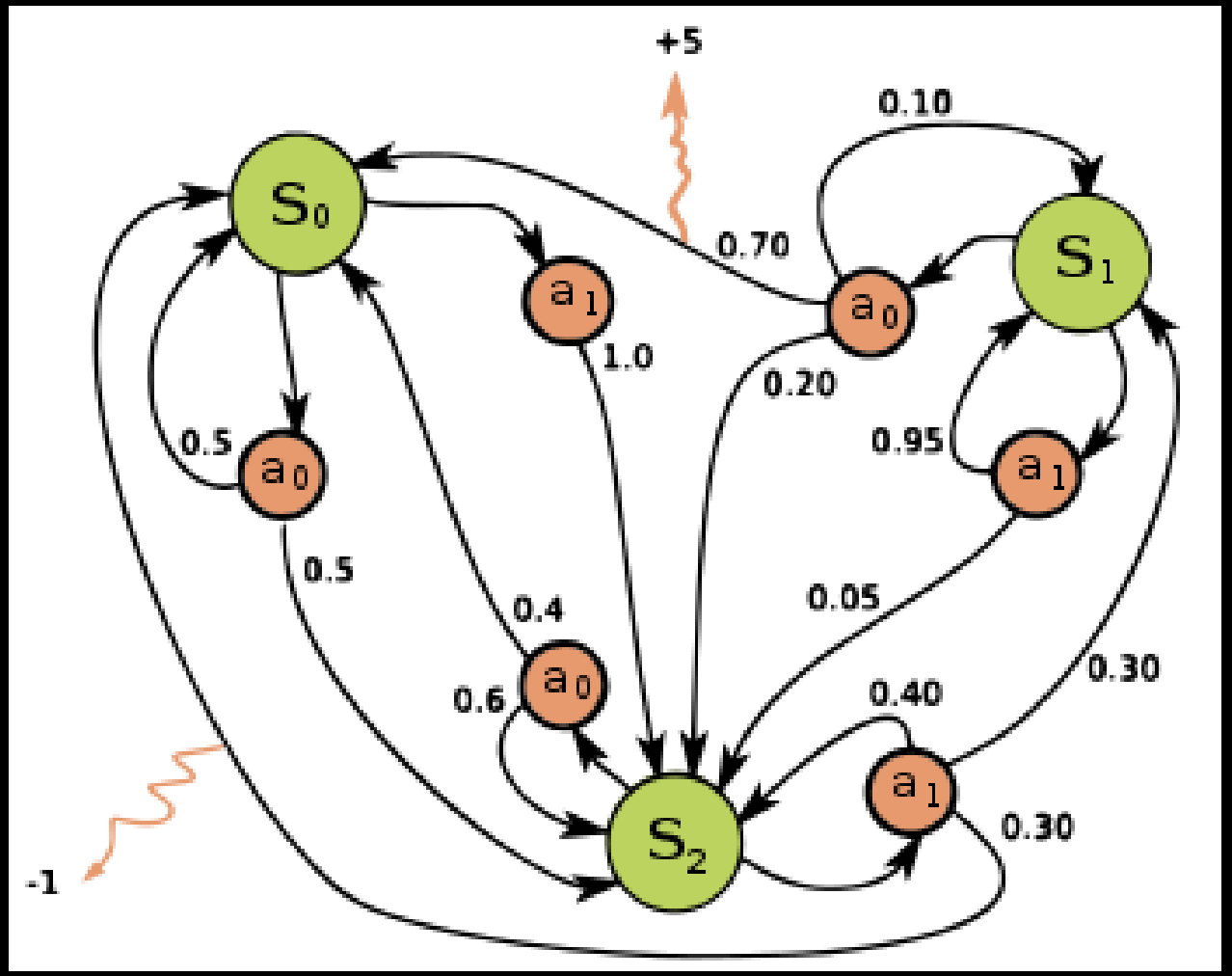
Theoretical Foundation of RL: Markov Decision Processes (MDP)

MDP consists of:

Markov process + Markov reward process + Actions

Actions

- At each time step, the chosen action influences what will become the next state
- Probability allows for randomness (e.g., turn car wheel to go right on icy patch but car slips and continues straight)



https://en.wikipedia.org/wiki/Markov_chain

Policy

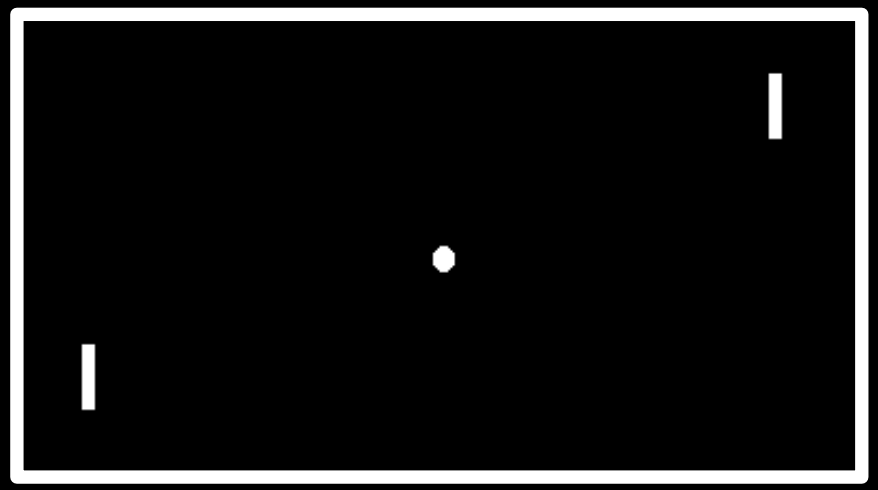
- Rules that dictate how an agent behaves
- Defined using a probability distribution over potential actions so there is randomness in the agent's behavior
- RL goal: find a good policy

Basic Ingredients for RL Methods

1. Observations of environment
2. Possible actions
3. Rewards

Basic Ingredients for RL Methods; e.g., Pong

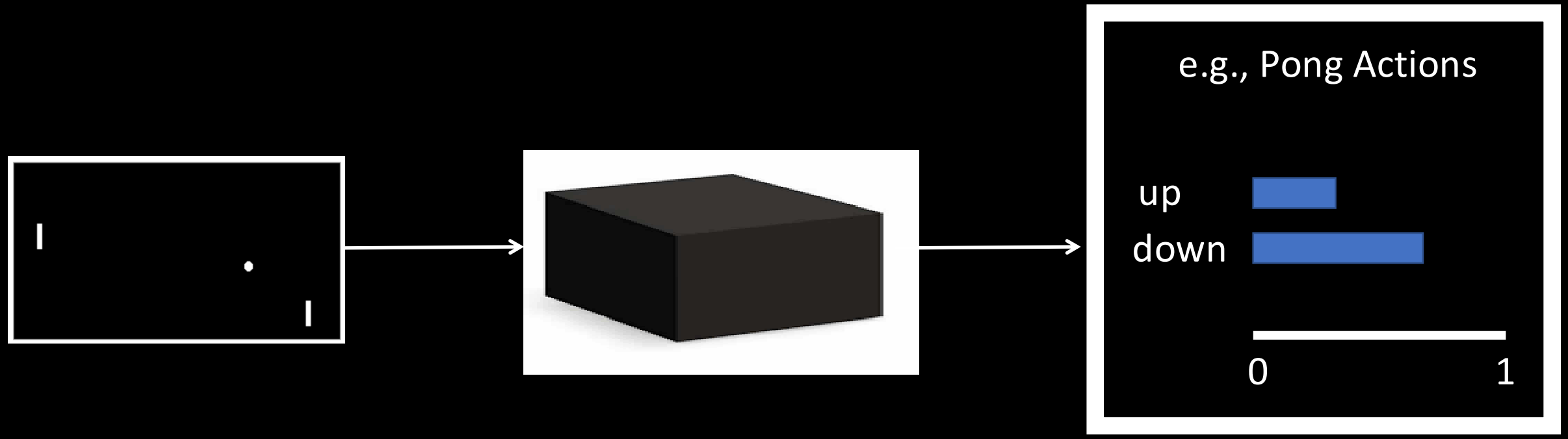
1. Observations of environment:



2. Possible actions: “up” and “down” paddle movements
3. Rewards: -1 if missed the ball; +1 reward if ball goes past opponent; 0 otherwise

Goal: Maximize rewards computing optimal “up” and “down” paddle movements

Policy Gradients: Approach

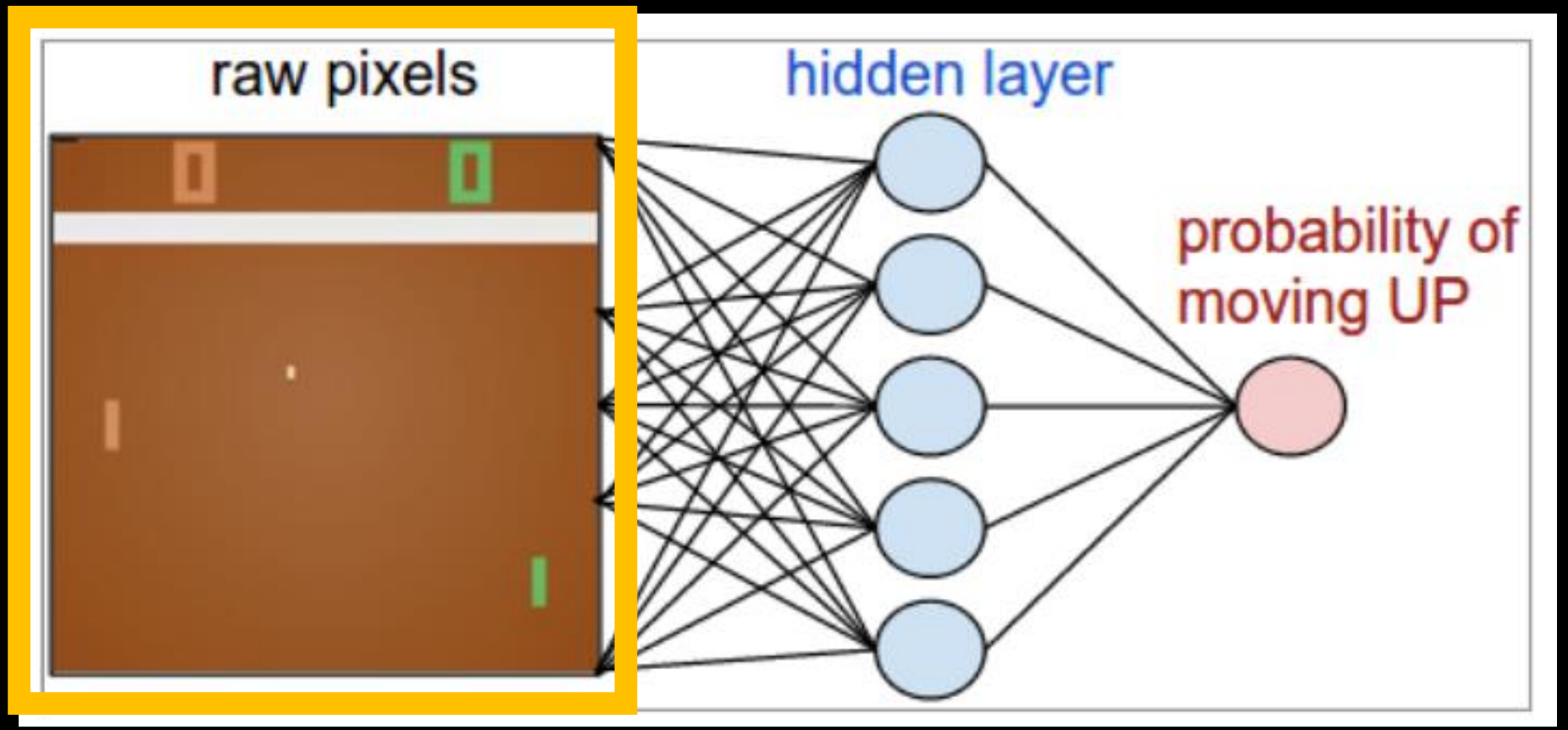


Policies (i.e., rules for agent behavior) represented with a probability for each possible action

Neural network trained to increase probability of actions leading to a good total reward and decrease probability of actions leading to a bad total reward

e.g., Learning Pong (2-layer NN with 200 hidden units)

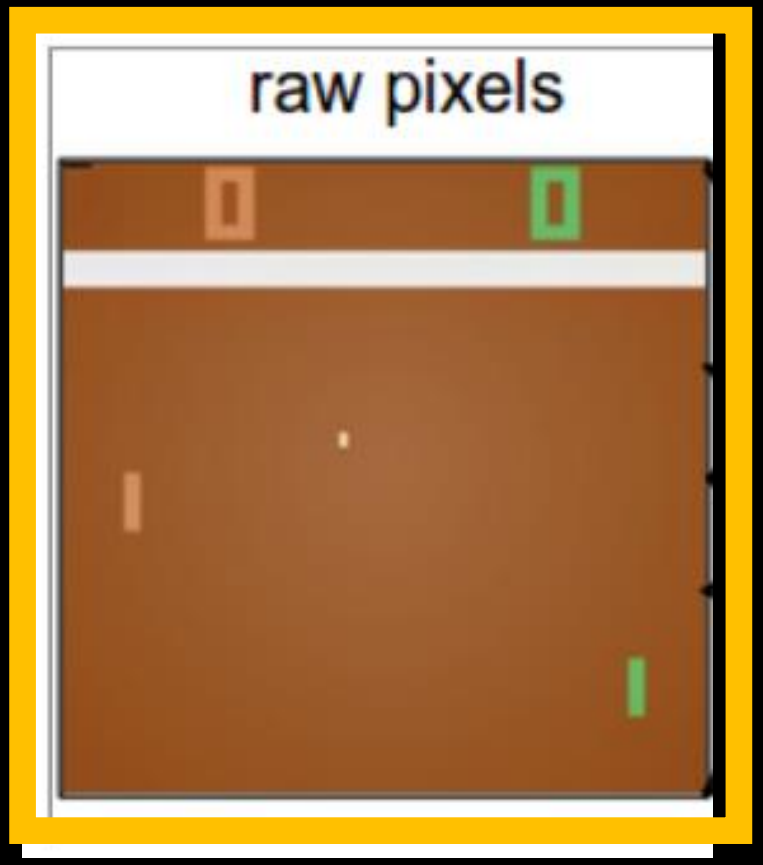
Given game state (as image), decide if to move paddle up (vs down)



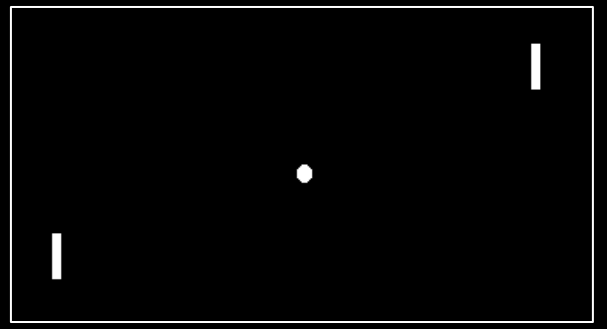
http://karpathy.github.io/2016/05/31/rl/

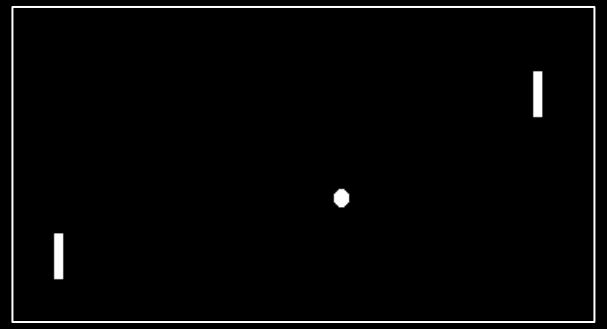
e.g., Learning Pong (2-layer NN with 200 hidden units)

How to capture motion in game state (i.e., image)?



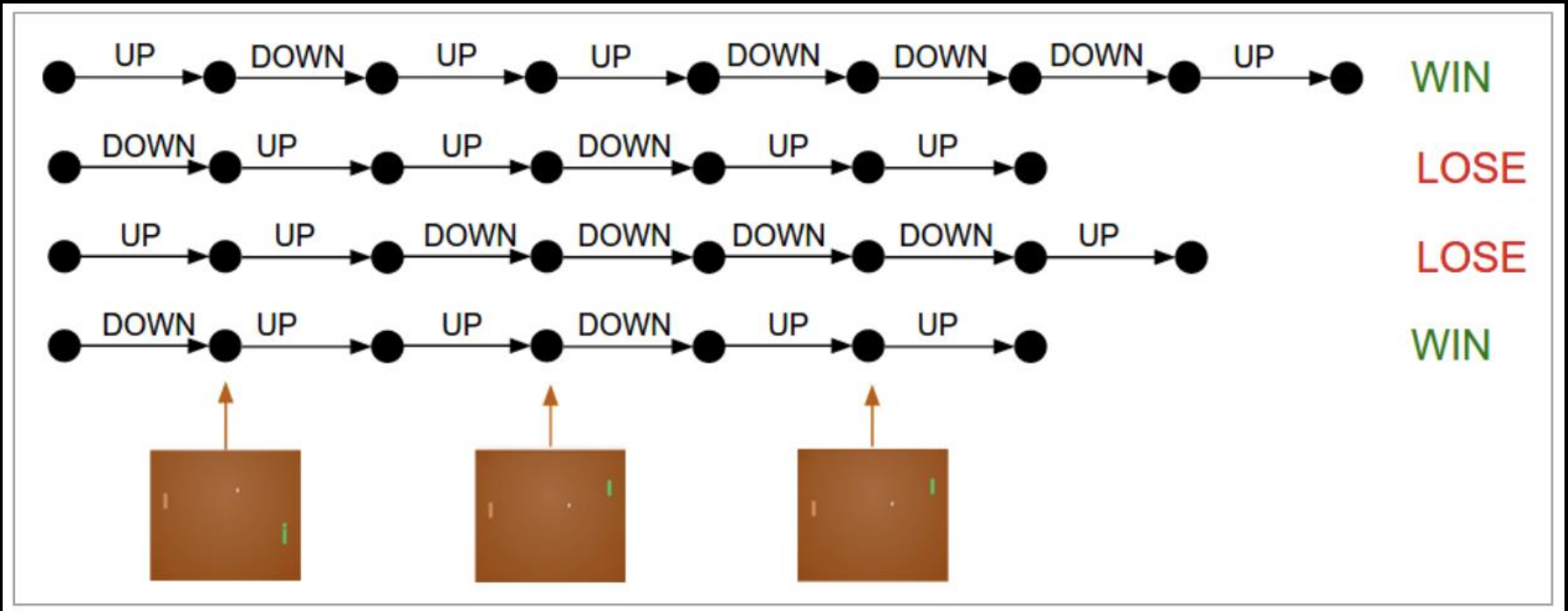
Use difference image
(i.e., subtract frame current from last frame)





http://karpathy.github.io/2016/05/31/rl/

e.g., Learning Pong: Training Protocol



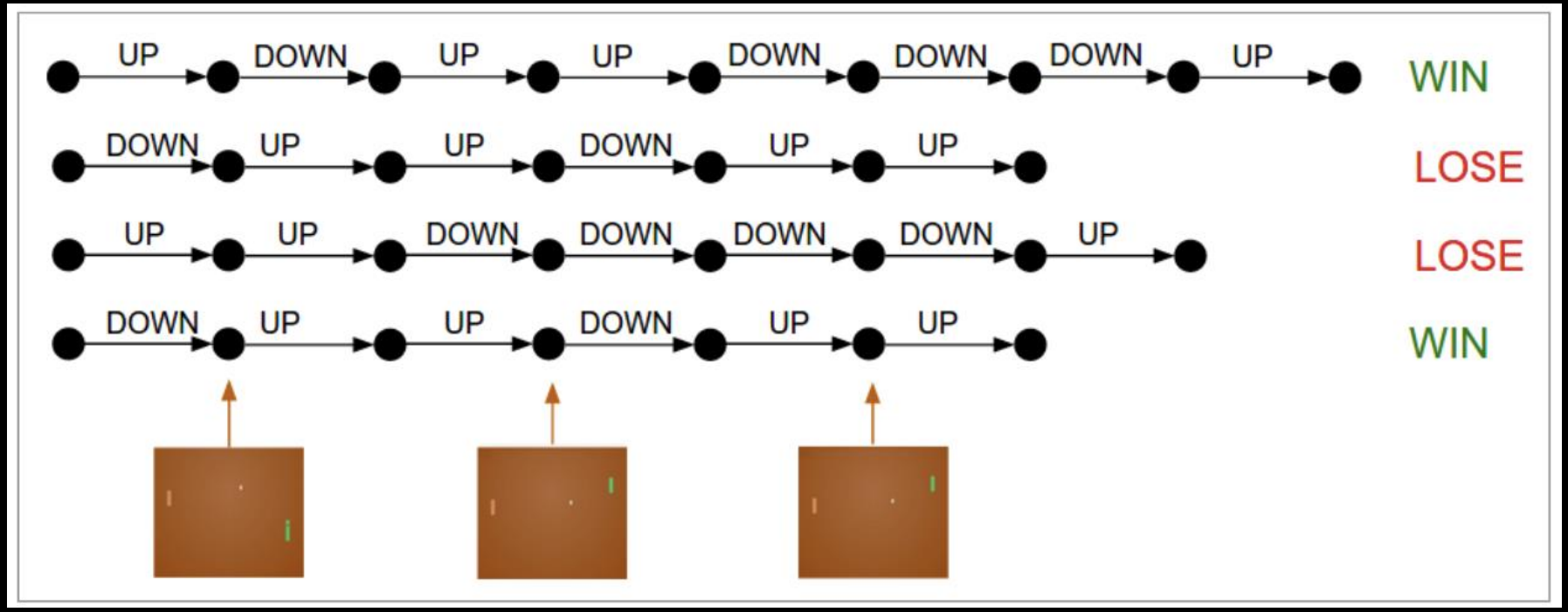
Assume 100 games played with 200 images/game

How many (action) decisions were made?

- 100x199

http://karpathy.github.io/2016/05/31/rl/

e.g., Learning Pong: Training Protocol

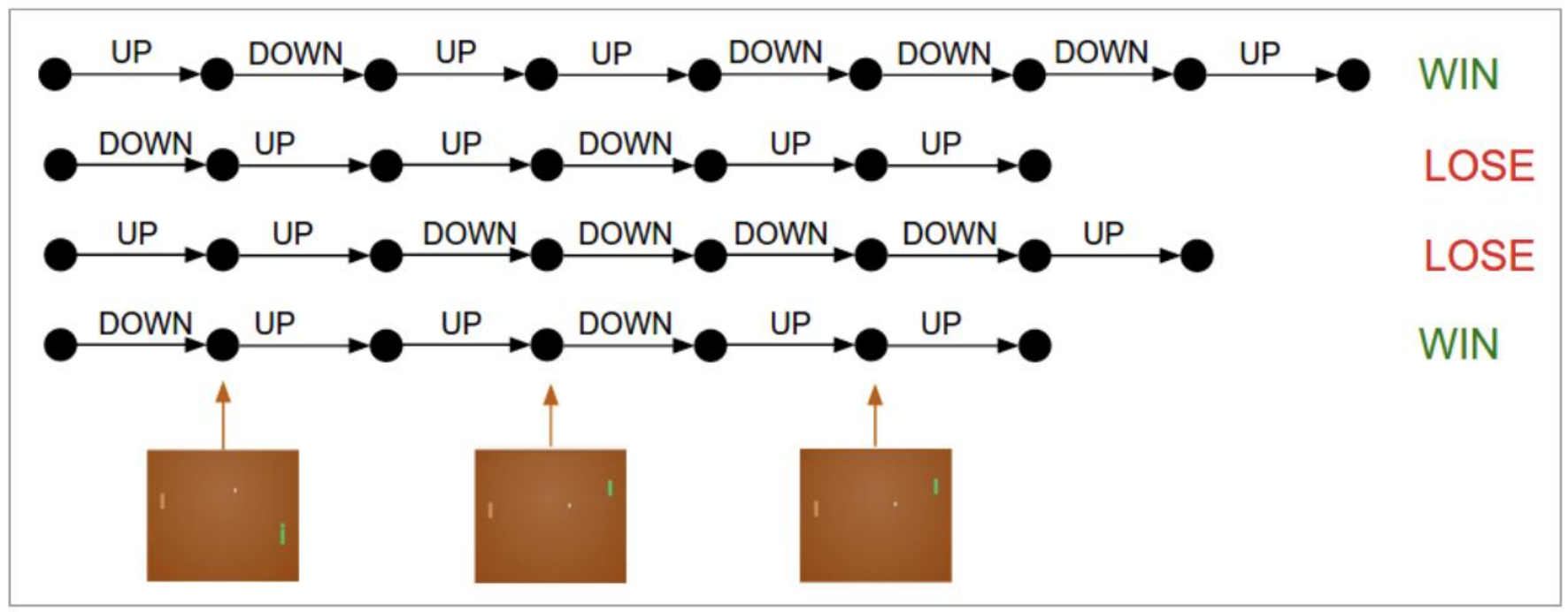


Assume 100 games played with 200 images/game; 12 games won & 88 lost

- How many winning decisions were made?
  - 2,388 (i.e., 12 x 199)
- How many losing decisions were made?
  - 17,512 (i.e., 88 x 199)

http://karpathy.github.io/2016/05/31/rl/

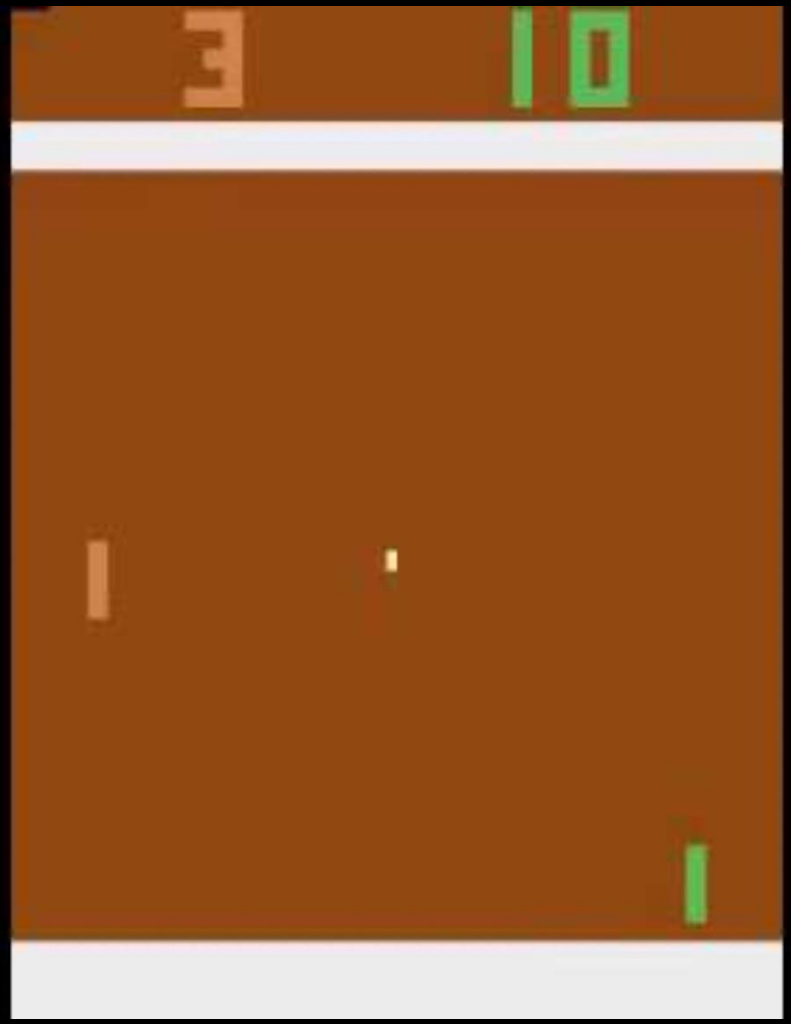
e.g., Learning Pong: Training Protocol



After 100 games, gradient updated to encourage actions that led to good outcomes (i.e., 2,388 winning up/downs) and discourage actions that led to bad outcomes (17,512 losing up/downs)

http://karpathy.github.io/2016/05/31/rl/

e.g., Pong Model: RL Model vs Pong's AI Model

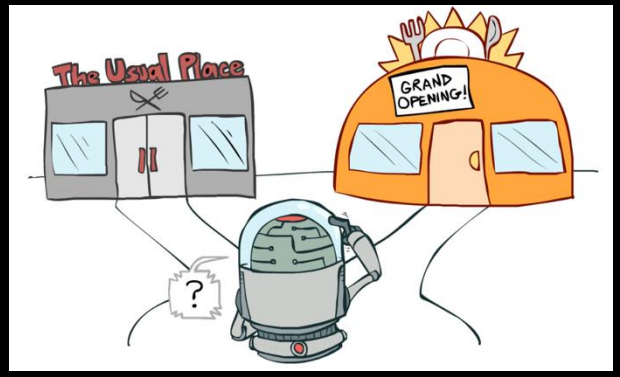


https://www.youtube.com/watch?v=YOW8m2YGtRg&t=16s

Why Reinforcement Learning is Difficult

- Agent must infer what it did wrong/right and so how performance can be maintained/improved based on (delayed) rewards; e.g., win/lose pong

- Agent needs to strike the appropriate balance between exploration and exploitation; e.g., order one's favorite food vs something new



https://huggingface.co/learn/deep-rl-course/unit1/exp-exp-tradeoff

- Agent must learn for non-verifiable tasks; e.g., human preferences, such as what is *funny* and what is *harmful*

Reinforcement Learning with Human Feedback

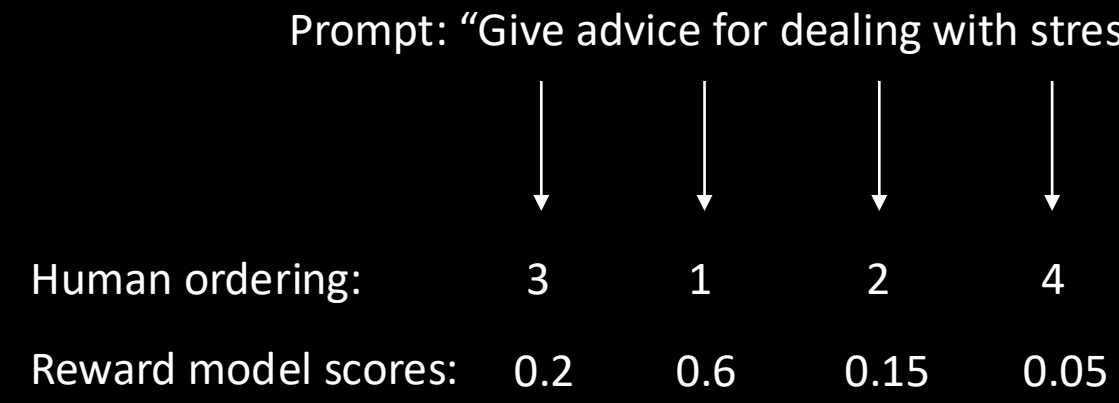
- Idea: train with most preferred solution; e.g.,

User: “My internet is not working right now”

Response	Human Preference Score
"Restart your router."	★★★
"Restart your router and check for service outages in your area."	★★★★
"Restart your router, check cables, and verify your service status. If that doesn’t work, contact support."	★★★★★

Reinforcement Learning with Human Feedback

- Idea: train with most preferred solution; human feedback address that reward functions are hard to specify (instead reinforces humans' preferences)
- Approach:
  1. Enable large-scale training by training a reward model to simulate humans' ranking preferences from a small amount of human input (using a ranking loss function)



  2. Fine-tune base model with reinforcement learning using the human-simulated preferences for the entire generated prediction (e.g., all predicted tokens in the sequence)

RLHF Challenge: Specification Gaming/Reward Hacking

- Example: training a robot to **clean a room,** with the reward provided for each piece of trash removed.
  - What is the problem with this learned behavior: break objects to increase the trash count
- How might this issue look with LLMs?
- Mitigating solutions:
  - Post-train only briefly with RLHF
  - Add loss term that encourages the model to stay close to original prediction distributions

Note: Conversational Tasks Bypass What is Supported in Many Evaluation Benchmarks

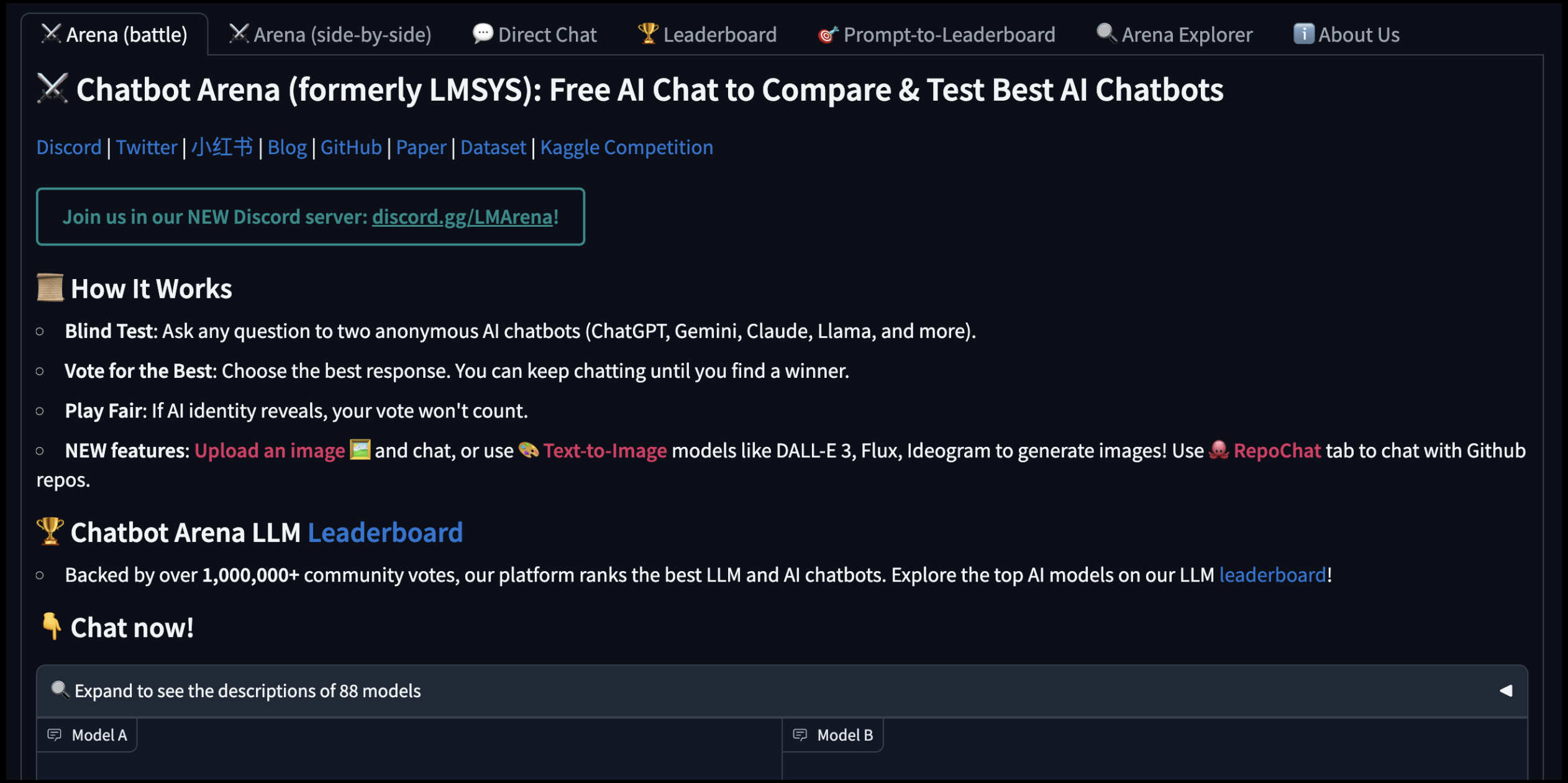


Commands

(brief with small vocabulary)

Conversations

(lengthy with large vocabulary)



https://lmarena.ai/

Want to Learn More About RL at CU?

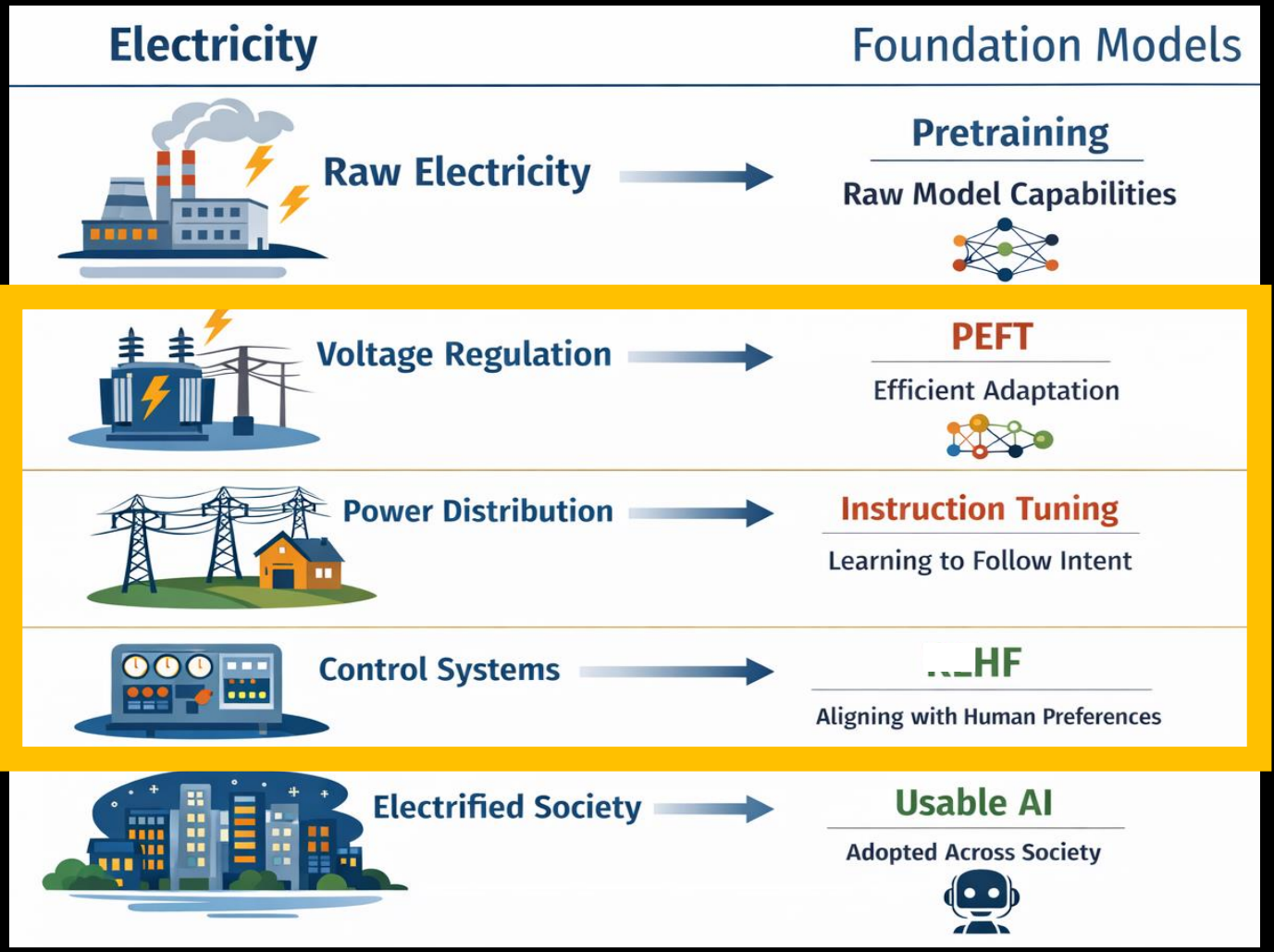
- Deep Reinforcement Learning — CSCI 7000
  - Alessandro Roncone
  - Typically offered in Fall

- Decision Making Under Uncertainty — ASEN 5264
  - Zachary Sunberg — https://github.com/CU-ADCL/CU-DMU-Materials
  - Typically offered in Spring

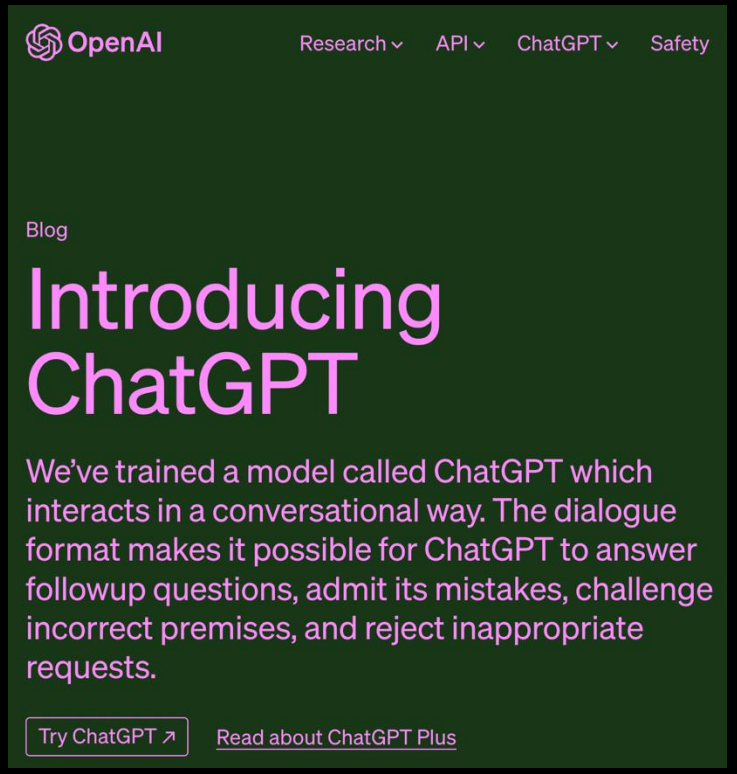
Outline

- The challenge: making foundation models useful
- Adapting models efficiently: parameter-efficient transfer learning
- Teaching models to follow human intent: instruction tuning
- Aligning models with human preferences: reinforcement learning from human feedback
- The shock: rapid societal adoption of foundation models

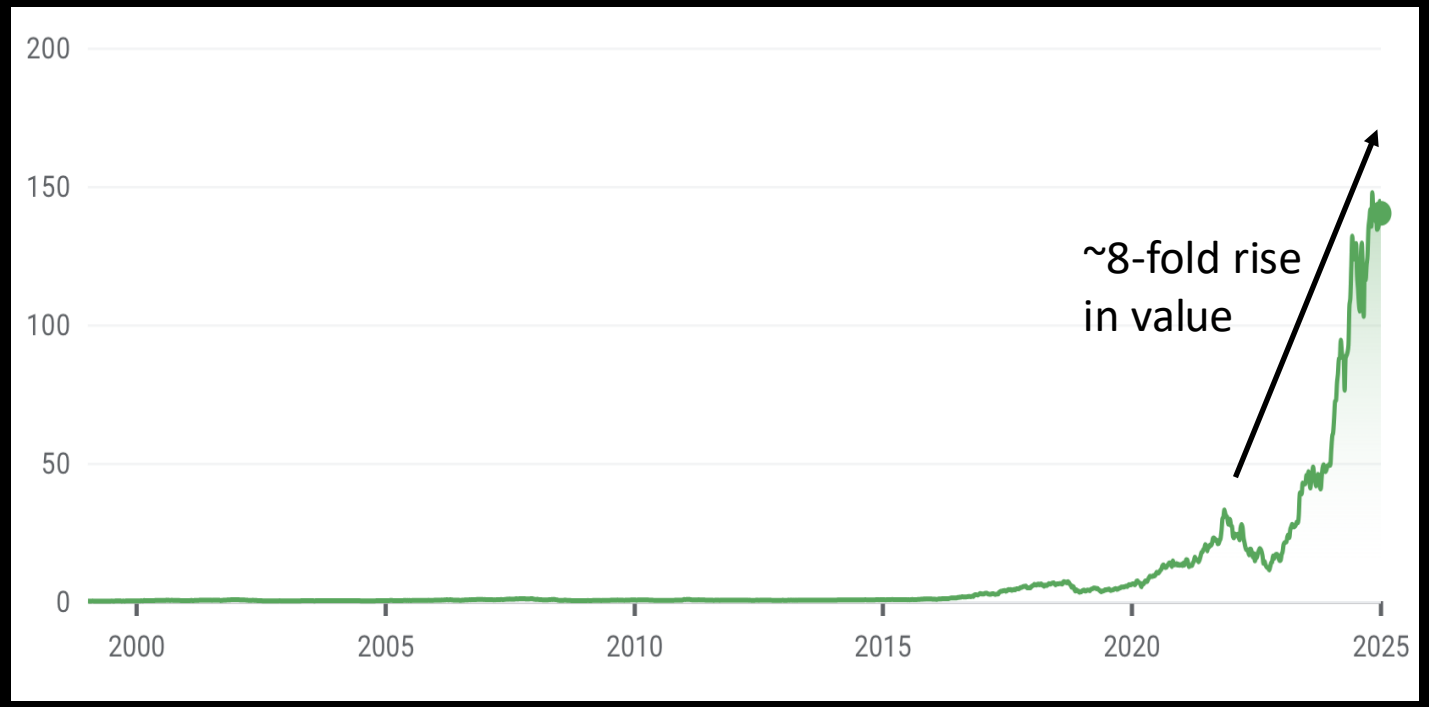
What Happened When ChatGPT (GPT 3.5) Bridged Raw Intelligence with Infrastructure?



Nov 2022: The Release of ChatGPT Changed the World's Expectations from AI…



- 1M+ users within five days
- ~10% of global adult population by July 2025

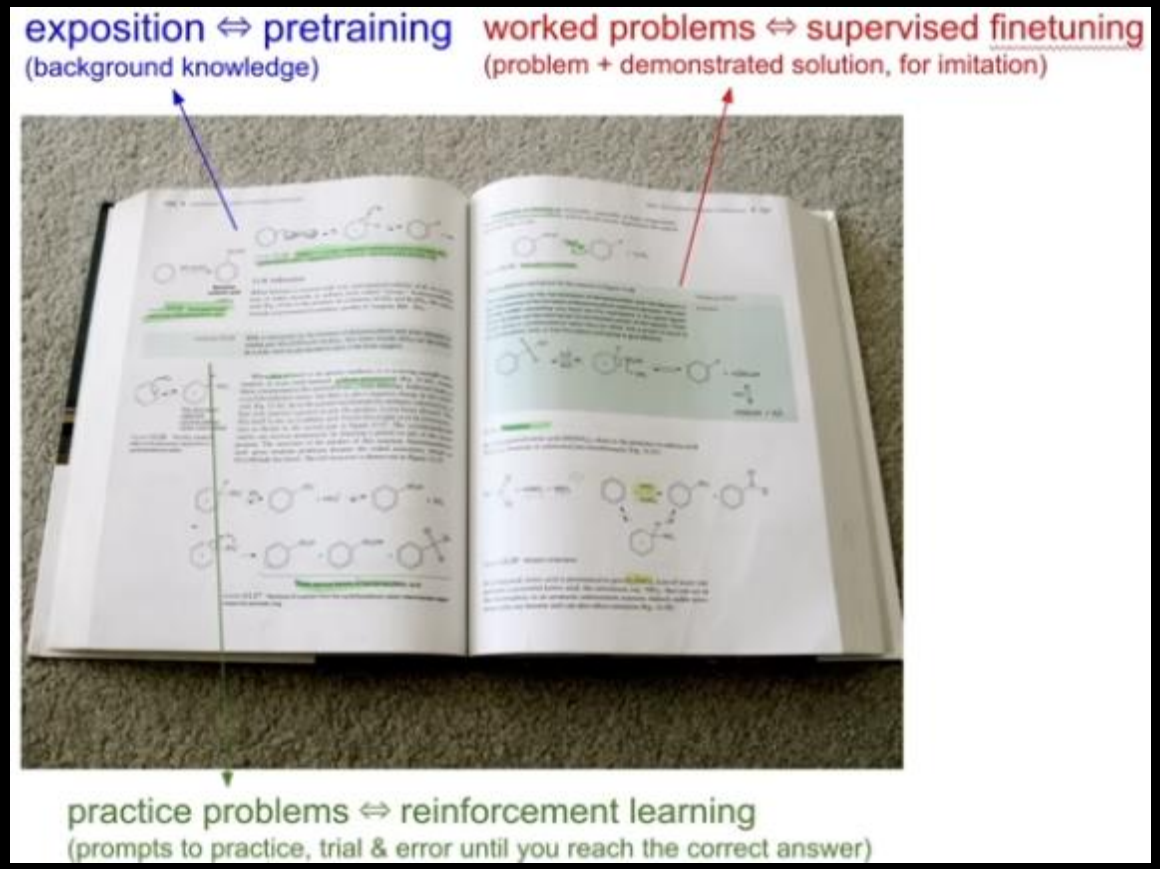


And helped NVIDIA, OpenAI's GPU provider, get much richer ☺
(market value of ~$4.3 TRILLION in March 2026)

https://cdn.openai.com/pdf/a253471f-8260-40c6-a2cc-aa93fe9f142e/economic-research-chatgpt-usage-paper.pdf
https://news.fsu.edu/news/education-society/2025/08/26/on-screen-and-now-irl-fsu-researchers-find-evidence-suggesting-chatgpt-influences-how-we-speak/

Debate: What Post-Training Steps to Include?

ChatGPT: instruction tuning + RLHF + hidden prompts



https://www.youtube.com/watch?v=7xTGNNLPyMI
(starts at approximately 2:12:00)

Llama extensions, such as Vicuna and Koala (2023): instruction tuning with LoRA

LLaMA: Open and Efficient Foundation Language Models

Hugo Touvron*, Thibaut Lavril*, Gautier Izacard*, Xavier Martinet
Marie-Anne Lachaux, Timothee Lacroix, Baptiste Rozière, Naman Goyal
Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin
Edouard Grave*, Guillaume Lample*

Meta AI

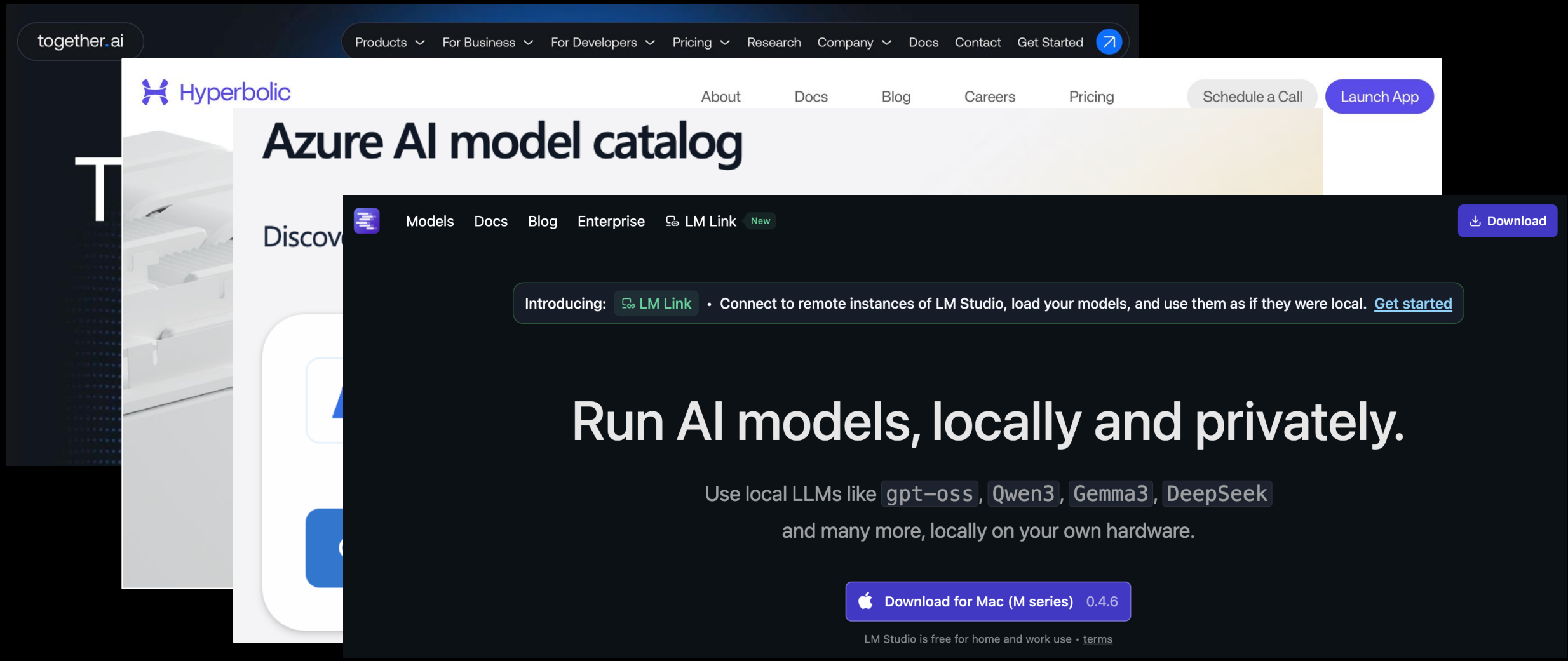
DeepSeek-R1 (2026): RLHF, with LoRA-based variants (permits going beyond human reasoning)

deepseek

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

To Help With Leveraging These Huge Models, Many Companies Provide Access to Them; e.g.,



Activity: Use What You Have Learned to Hypothesize and Analyze Models' Behaviors

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The End